Guillain-Barré syndrome

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Primer

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Guillain–Barré syndrome

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Abstract

Guillain–Barré syndrome (GBS) is a rare immune-mediated polyradiculoneuropathy. Patients typically develop rapidly progressive weakness and sensory deficits that can result in complete paralysis requiring mechanical ventilation. GBS is usually a monophasic disease in which an aberrant immune response to an infection or other trigger damages the peripheral nerves. For example, in patients with preceding *Campylobacter jejuni* infection, molecular mimicry causes a cross-reactive antibody response to nerve gangliosides. Diagnosis is based on clinical features, supported by cerebrospinal fluid analysis and nerve conduction studies. Effective treatments include plasma exchange and intravenous immunoglobulins. However, ~20% of patients who received treatment are unable to walk after 6 months and ~5% die as a consequence of GBS. Important knowledge gaps in GBS include its pathogenesis, especially after viral infections. In addition, there is a lack of specific biomarkers to improve the diagnosis, monitor the disease activity, and predict the clinical course and outcome of GBS. Major challenges for the future include finding more effective and personalized treatments, which are affordable in low-income and middle-income countries, and preparation for outbreaks of infections as potential triggers for GBS.

Sections

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Introduction

Guillain–Barré syndrome (GBS) is a rare but potentially life-threatening immune-mediated polyradiculoneuropathy. The syndrome was named after George Guillain and Jean Baptiste Barré who, together with André Strohl in 1916, published a seminal report on cerebrospinal fluid (CSF) findings in patients with acute flaccid paralysis that were distinctive from poliomyelitis[1]. GBS is a typical post-infectious disorder, triggered by bacterial or viral infections, for example, infection with *Campylobacter jejuni* or Zika virus, explaining the occasional link with outbreaks or epidemics[2,3].

GBS typically presents with paraesthesia, pain or weakness of the limbs that rapidly progresses in days. Around 20% of patients may develop weakness in all four limbs (tetraparesis) and respiratory failure requiring mechanical ventilation[4]. GBS may also affect cranial nerves and induce facial palsy, ophthalmoplegia, or bulbar weakness. The typical disease course of GBS comprises an initial progressive phase, followed by a plateau and then recovery (Fig. 1). The predominant phenotype is motor-sensory, but patients may also present with a motor form of GBS, without sensory deficits, or a clinical variant such as Miller Fisher syndrome (MFS) (Fig. 2 and Table 1). Early diagnosis may be challenging because of the diversity in clinical presentation. Examining the CSF is recommended for cell count, protein level and specific tests to exclude other disorders. Nerve conduction studies (NCS) usually demonstrate evidence of peripheral nerve involvement. The only two proven effective treatments for GBS are intravenous immunoglobulins (IVIg) and plasma exchange. Despite these treatments, patients may still die of GBS and a substantial proportion of patients experience residual limitations or complaints. Even if GBS usually has a monophasic disease course, 2–5% of patients relapse even years after the first disease episode[5–7].

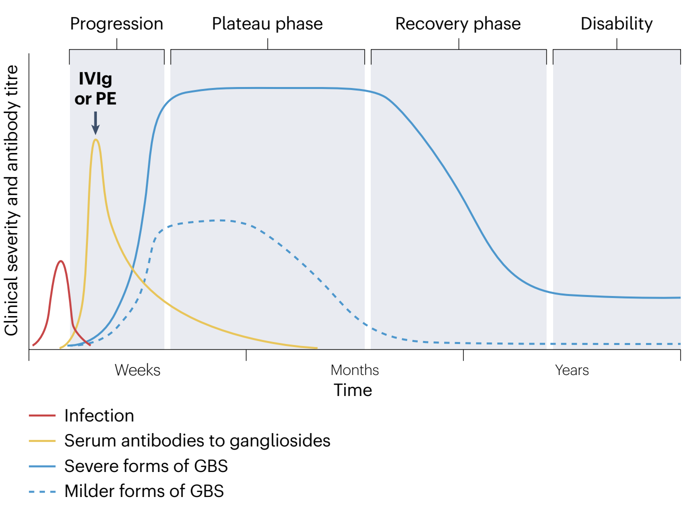


Fig. 1 | Disease course of GBS. Typical monophasic disease course of Guillain–Barré syndrome (GBS), characterized by an initial acute and progressive phase of <4 weeks, followed by a plateau phase and recovery phase of weeks to years and residual disability. GBS is usually preceded by a mild or subclinical infection that induces a transient antibody response to gangliosides, which damages the peripheral nerves and results in neurological deficits. Plasma exchange (PE) and intravenous immunoglobulins (IVIg) are equally effective treatments in the acute stage of GBS. Patients show a remarkably diverse clinical severity, course and outcome.

In this Primer, we provide an overview of the current understanding of GBS epidemiology and mechanisms as well as of its diagnosis, outcome prediction and management, including current challenges. The Primer includes considerations related to the 2023 publications of the GRADE-based guideline for GBS from the European Association of Neurology (EAN) and Peripheral Nerve Society (PNS)[8,9] as well as the International GBS Outcome Study (IGOS), a prospective cohort study on disease course and outcome[10]. Further research is required to find more effective and personalized treatment options and to be prepared for future outbreaks of infections as potential triggers for GBS. This requires a global health approach that includes low-income and middle-income countries (LMICs).

Epidemiology

Global incidence and prevalence

GBS is the most common cause of acute flaccid paralysis worldwide[11]. The annual global incidence is estimated at 1–2 per 100,000 persons per year and is mainly based on data from studies conducted in North America and Europe[12,13]. In most LMICs, including large parts of the African and Asian continent, no epidemiological data are available[12,14] (Fig. 3). Males are affected ~1.5 times more frequently than females[12,13]. The reason for the male predominance in GBS is not known[12]. Although all age groups can be affected, the risk of GBS steadily increases by 20% for every 10-year increase in age[12]. In ageing populations in North America and Europe, the reported incidence is highest in those over the age of 60 years[12]. However, in LMICs with younger populations, GBS is most frequently reported in patients 20–40 years of age[4,14–17].

The incidence of GBS fluctuates seasonally, probably related to the infections that can trigger the disease[18]. In the northern hemisphere, increased incidence in winter months (January–March) in western, middle eastern and far eastern countries is thought to be due to increased frequency of respiratory tract infections. By contrast, in China and the Indian subcontinent, increased incidence has been reported during summer months (June–August), which is thought to be due to an increased frequency of diarrhoeal illnesses[19,20].

Mortality

GBS mortality in most studies varies between 2% and 10% and differs depending on region. In Europe and North America, reported mortality rates are 2–7%, whereas rates of 14–17% and 16% have been reported for Bangladesh and Egypt, respectively[4,21–23]. These disparities between higher-income and lower-income countries are likely due to the limited availability of treatment or specialized care in lower-income countries. For instance, in Bangladesh, only an estimated 10–12% of patients can afford IVIg or plasma exchange[14]. Risk factors for death include high age, severe weakness at an early stage of disease, late presentation and mechanical ventilation[24].

Preceding events

Several infections have been associated with GBS in case–control studies, including infection with *C. jejuni*, *Mycoplasma pneumoniae*, cytomegalovirus, Epstein–Barr virus, hepatitis E virus or Zika virus[2,3,25–27]. *C. jejuni*, a bacterium causing diarrhoeal illness, is the most frequent preceding infection globally, occurring in ~30% of patients with GBS[28]. *M. pneumoniae* causes a respiratory tract infection mostly in children and is the second most frequent infection preceding GBS, occurring in ~10% of patients[28]. Infection with several other pathogens, including dengue virus, chikungunya virus, varicella zoster virus and human immunodeficiency virus, as well as vaccinations, malignancy, trauma

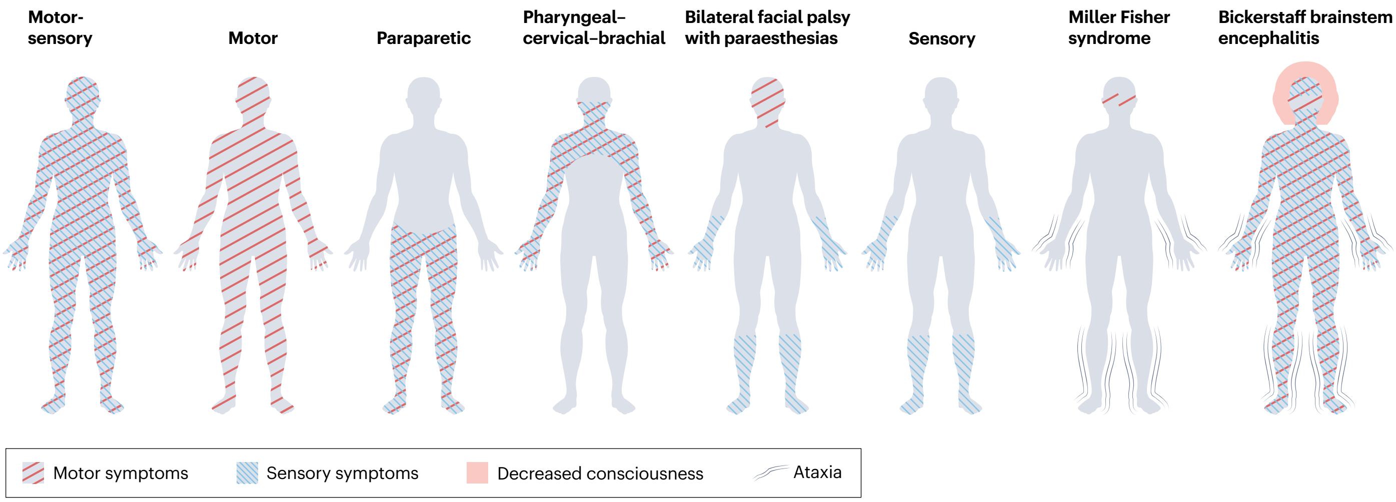


Fig. 2 | Clinical patterns of GBS. Guillain–Barré syndrome (GBS) has a highly variable clinical presentation and course, and several distinct forms and variants can be identified. Some patients may have incomplete variants, overlapping variants or progress from one form to another, reflecting the spectrum of GBS manifestations. These forms and variants have a varying incidence and the associated serum antibodies have a varying specificity to gangliosides[126] (Table 1). Adapted from ref. 129, Springer Nature Limited.

and surgery, specifically procedures on bones and digestive organs, have been linked to GBS in uncontrolled case series or surveillance studies[29–36]. Conclusive evidence for the relation between GBS and other reported preceding events, such as pregnancy and allergy, is lacking.

Outbreaks of infectious disease. Although GBS is a sporadic disease, an increased incidence of GBS has been reported following epidemics of infectious agents. In 1995, clusters of patients with GBS and preceding diarrhoea were linked to a local outbreak of *C. jejuni* infection in northern China[19]. Subsequently, several other GBS clusters have been linked to epidemics of *C. jejuni* in China (2007), Mexico (2011) and Peru (2018–2023)[37–41]. The risk of GBS after *C. jejuni* infection has been estimated to be 100 times higher than the background incidence of GBS[42,43].

Several studies have found an association between influenza-like illness and GBS[44,45]. During the H1N1 pandemic in 2009, a surveillance study among children reported a noticeable increase in GBS just after the peak of the H1N1 pandemic. Of the 57 children diagnosed with GBS, 49 had clinical or laboratory evidence of influenza infection in the 3 months before the onset of neurological symptoms[46].

The incidence of GBS was also increased during the Zika virus epidemic in South America in 2015–2016, and epidemiological and case–control studies have provided evidence of a true association between this infection and GBS[3,47–49]. A meta-analysis estimated the overall risk of reported GBS at 2.0 (95% CI 0.5–4.5) per 10,000 Zika virus infections, which is 10–20 times higher than the annual global incidence of GBS, which is estimated at 1–2 per 100,000 persons per year[50]. Other arbovirus infections, including dengue and chikungunya fever, have been linked to GBS during outbreaks but these associations are less well studied[51–53].

The SARS-CoV-2 pandemic is the most recent infectious disease outbreak purportedly linked to GBS. Some studies reported an increase in GBS hospital admissions during the pandemic[54,55], and a large meta-analysis indicated a very small increased risk of GBS after SARS-CoV-2 infection[56]. However, these studies are confounded by ascertainment bias and lack of diagnostic verification inherent in the use of hospital coding data[57]. By contrast, large epidemiological studies found a decrease in GBS incidence compared with previous years, although this decreased incidence is probably due to reduced exposure to other pathogens following hygienic measures and curfew[54–56,58,59]. Overall, it is difficult to refute or confirm a causal association between COVID-19 and GBS. There is no evidence that GBS after COVID-19 is related to a specific clinical phenotype[60]. Based on the available data, it is clear that, if SARS-CoV-2 infection causes GBS, it does so very infrequently.

Vaccines. Vaccines were first linked to GBS in the late 1970s when a 7.3-fold increase in risk of GBS was reported following the H1N1 influenza ('swine flu') vaccine campaign in the USA[61]. The potential link between GBS and other vaccines has been thoroughly studied. For most of these vaccines, no association with the development of GBS exists. For some vaccines, including other seasonal influenza vaccines, measles and measles–mumps–rubella (MMR) vaccines, human papillomavirus vaccines, and quadrivalent meningococcal vaccine, there may be an increased risk of <1 GBS case per one million vaccinated persons[32,62]. Furthermore, in the case of influenza, several studies have shown that the incidence of GBS is lower in vaccinated compared with unvaccinated groups[45,63]. This suggests an overall protective effect of influenza vaccination from GBS at the population level, mediated through the prevention of influenza infection.

The SARS-CoV-2 vaccines are the most recently debated vaccines in relation to GBS. Several large cohort studies, using data from millions of vaccinated individuals, reported a probable association between the adenovirus-vectored vaccines and GBS[64]. In most studies, the increased risk was around 5 cases per 1 million vaccinated individuals[64]. For other types of SARS-CoV-2 vaccines, such an association has not been found[64]. Some studies indicated that vaccination for COVID-19 may even reduce the risk of developing GBS[56,65].

Table 1 | Features of Guillain–Barré syndrome clinical patterns

Clinical pattern	Motor-sensory	Motor	Paraparetic	Pharyngeal–cervical–brachial	Bilateral facial palsy with paraesthesias	Sensory	Miller Fisher syndrome	Bickerstaff brainstem encephalitis
Percentage of patients	30–85%	5–25%	5–10%	<5%	<5%	<1%	4–25%	<5%
Presence of flaccid paralysis	Yes	Yes	Lower limbs	Proximal upper limbs	No	No	With overlap	With overlap
Altered sensation	Yes	No	Lower limbs	No	Distal limbs	Yes	With overlap	With overlap
Cranial neuropathy	Yes	Yes	No	Bulbar	Facial	No	Ocular motor	Ocular motor
Central nervous system involvement	No	No	No	No	No	No	No	Altered consciousness, hyper-reflexia
Associated antibodies (IgG)	–	GM1, GD1a	–	GT1a	–	GD1b	GQ1b, GT1a	GQ1b, GT1a

Includes data from ref. 126.

In general, the benefits of currently recommended vaccines far outweigh the hypothesized risk of developing GBS or a relapse in patients with a previous episode of GBS. Immunization may even reduce the risk of developing GBS as it prevents infections that can trigger GBS.

Mechanisms/pathophysiology

Overview

GBS is usually a monophasic condition in which the immune system aberrantly disrupts the function and structure of the peripheral nervous system. GBS seems to be triggered by a preceding infection, vaccination or, rarely, another immunological stimulus such as surgery. However, the mechanisms by which the triggering event leads to a failure of immunological self-tolerance, the relevant host risk factors, the immune effectors driving nerve injury, their antigenic targets and how self-tolerance is ultimately restored are incompletely defined. Although generally labelled as an inflammatory neuropathy, it is also uncertain whether GBS is truly auto-inflammatory (driven by innate immune mechanisms) rather than autoimmune (driven by adaptive immunity). It is clear that GBS does not have the features of a typical autoimmune disease. Its male preponderance, short duration, self-resolving monophasic disease course, lack of associations with specific human leukocyte antigens[66,67] and other autoimmune diseases, and lack of response to corticosteroids[68] mark it out as atypical in this context. Furthermore, whether the clinically defined syndrome is produced by a single or diverse range of underlying pathophysiological processes is uncertain.

Electrophysiology has historically been used to divide GBS into demyelinating and axonal forms, taken to indicate distinct pathological processes targeting the myelin sheath or axon, respectively. However, mixed axonal–demyelinating, equivocal or even normal NCS findings are frequent, suggesting a complex electrophysiological involvement that mirrors the diversity of pathological findings[69]. Regardless of the primary mechanism, the final outcome in GBS is peripheral nerve damage that includes endoperineural and subperineurial oedema, inflammatory infiltrates, segmental demyelination, and axonal damage[70–72].

Antibodies

The presence of inflammatory oedema in the absence of cell infiltrates (particularly in early stages of the disease), the deposition of complement and IgG and IgM in nerve specimens[73] and, most importantly, the response to plasma exchange[74] made the search for a pathogenic humoral factor in blood or CSF one of the leading research topics in GBS.

Although autoantibodies targeting a range of peripheral nerve antigens have been described (Box 1), the association with and pathogenic role of ganglioside antibodies in GBS is supported by the most robust evidence. Gangliosides are a family of glycosphingolipids formed by a ceramide core and a variable number of sialic acid residues. Although gangliosides are present in plasma membranes of all tissues, certain ganglioside species are enriched in specific parts of the peripheral nervous system. Ganglioside antibodies were first identified in the sera of patients with GBS in the 1980s[75,76]. Since then, their mechanisms of induction, clinical associations and pathological effects have been extensively studied[77].

Although the presence of ganglioside antibodies has been reported following several different infectious triggers, the most comprehensively defined GBS pathophysiological process is that which follows *C. jejuni* infection (Fig. 4). In this paradigm, some individuals, who may have an intrinsically increased innate immunity response to *C. jejuni*[78], develop antibodies against bacterial lipo-oligosaccharides (LOS) that cross-react with peripheral nerve gangliosides through molecular mimicry mechanisms. A study using an array of LOS ganglioside mimics to explore the affinities and specificities of GBS-associated antibodies further supports the concept that these antibodies are induced by the bacterial LOS but cross-react with lower affinity to host gangliosides[79]. Anti-ganglioside antibodies in GBS are usually IgG but IgM and IgA anti-ganglioside antibodies may also be present[76,80,81]. Some of these antibodies may bind preferably to a complex of two different gangliosides rather than to either one of these gangliosides. Their levels are typically at a maximum in the acute disease phase (within the first 4 weeks of onset), then decline progressively and, most of the time, disappear within 6 months[82,83]. Persistently elevated GM1 antibody titres may be associated with poor outcome and ganglioside antibody levels may also spike again in recurrent disease[82].

C. jejuni with monofunctional (a2,3) sialyltransferase Cst-II (Thr51/Asn51) activity express sialylated surface LOS and induce antibodies to gangliosides such as GM1 and GD1a, whereas those with bifunctional (a2,3 and a2,8) sialyltransferase Cst-II (Asn51) activity express disialylated LOS and induce antibodies to gangliosides such as GT1a and GQ1b[84]. The resulting antibody specificity influences the clinical phenotype due to specific ganglioside enrichment at different peripheral nervous system sites (Fig. 2, Fig. 4 and Table 1). Neuropathogenic effects occur when these antibodies target gangliosides at the motor nerve terminal[85] and/or node of Ranvier[86]. Complement activation at these sites causes initial peripheral nerve dysfunction followed by structural injury, which may include axonal degeneration, segmental demyelination and/or disruption of the nodal–paranodal architecture[86].

Both in vivo and in vitro, the presence of additional gangliosides or other glycolipids can either enhance or suppress antibody binding. This has been most clearly demonstrated in experimental mouse models showing that the presence of GD1a prevents the binding of some but not other GM1-specific antibodies to GM1 in neural membranes[87]. These effects are hypothesized to occur due to the formation of heteromeric glycolipidic complexes, meaning that antibody binding is not correlated with the tissue distribution of any one ganglioside in an easily explicable way. A further level of complexity is added by the fact that, at certain sites (notably the motor nerve terminals), ganglioside antibodies can be rapidly internalized and removed from the circulation by endocytosis through an antigen-dependent mechanism further modifying their pathological effects, at least in mouse models[88,89]. Although defective internalization leading to autoantibody accumulation might explain general or localized susceptibility to GBS pathology, there is no direct evidence for this in patients.

Ganglioside antibodies binding at the motor nerve terminals in mouse ex vivo preparations induce a massive quantal release of acetylcholine, and a sharp spike in the miniature endplate potential frequency, followed by conduction failure[90,91]. Antibody binding to the nodal axolemma similarly cause conduction block due to nodal and paranodal membrane disruption. In both cases, these effects are due to activation of the complement cascade and influx of calcium through the terminal complement complex (also known as the membrane attack complex, C5b-9). Unchecked, this leads to the activation of calpains and caspases and axonal degeneration. According to data from mouse models, this occurs, at least in part, due to the calpain-mediated proteolysis of neurofilaments, cytoskeletal proteins, such as ankyrin G, and sodium channels[85,86].

Although initially associated with primary axonal variants of GBS, studies in transgenic mice in which ganglioside expression is limited to Schwann cell membranes have shown that GM1 ganglioside antibodies can also target the glial paranode[92]. Not only does this also induce conduction failure but complement-mediated Schwann cell injury is followed by axolemmal nanoruptures, which then facilitate (non-membrane attack complex related) calcium influx leading to secondary axonal degradation[70]. These observations further blur the distinction between demyelinating and axonal GBS and suggest a possible final common pathway for axonal degeneration.

Histological studies

In 1949, authors of a clinicopathological report on 50 fatal cases of GBS attempted to recreate the pathophysiological sequence of events by studying autopsy-derived peripheral nerve tissue at different stages of disease. The crucial observation was an initial pathological change

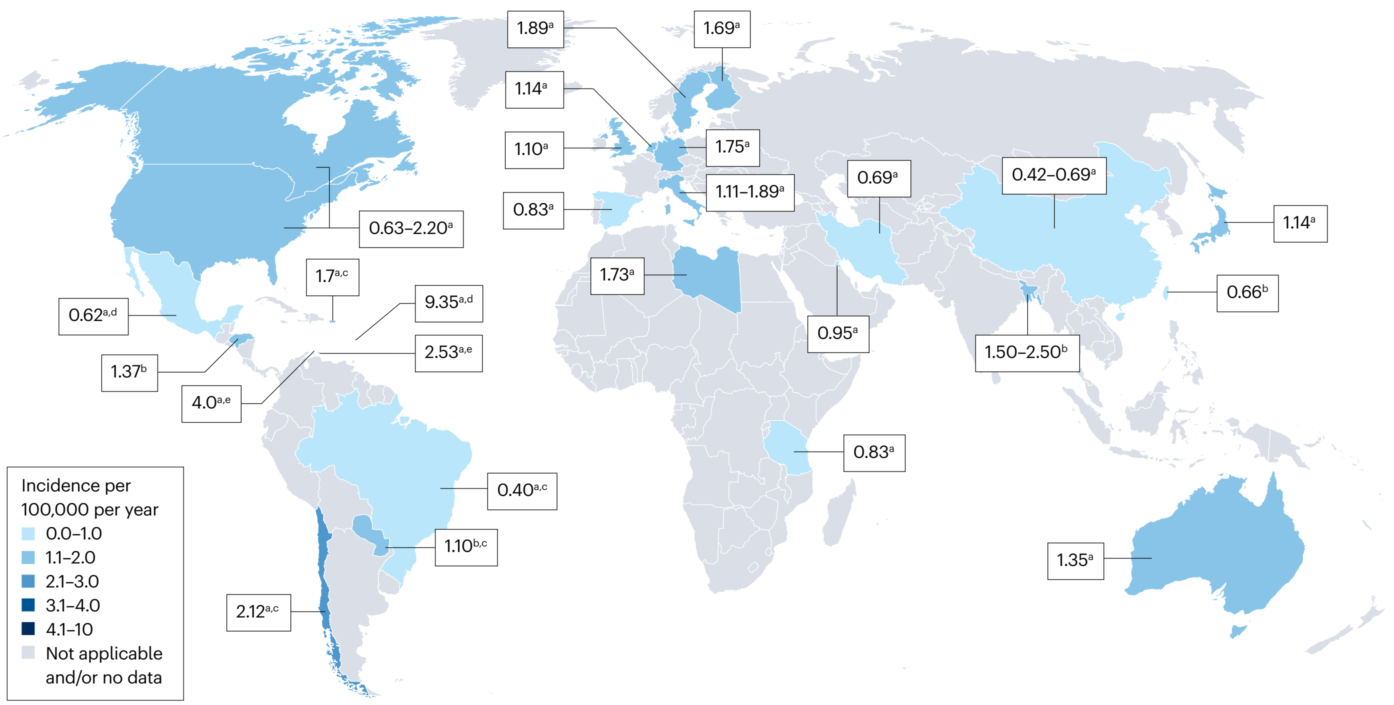


Fig. 3 | Global epidemiology of GBS. This map shows an overview of the worldwide incidence of Guillain–Barré syndrome (GBS) in different regions and in children and adults. The highlighted countries have a reported incidence, which is shown in the boxes. Most data on the incidence of GBS is from populations in high-income countries and only few studies have included populations from low-income countries. The reported incidence of GBS in general ranges from 0.16 to 3.0 cases per 100,000 persons per year. If a study was performed specifically in children, a lower incidence rate than in adults is to be expected. If a study was performed in relation to *Campylobacter jejuni* outbreaks or the Zika virus epidemic, a higher incidence rate than without these outbreaks or epidemics is to be expected. Data for Africa from refs. 15,212; data for Asia from refs. 20,213–215; data for Central America from ref. 216; data for Europe from refs. 12,217–219; data for the Middle East from refs. 220,221; data for North America from refs. 12,13,222; data for Oceania from ref. 223; data for South America from refs. 222,224–226. [a]All age groups. [b]Children. [c]Before the Zika virus epidemic. [d]During the Zika virus epidemic. [e]During a *C. jejuni* outbreak.

Box 1 | Antibodies in GBS

Antibodies to ganglioside are found in 20–70% of patients with Guillain–Barré syndrome (GBS)[19,22,75–77,82,83,138,139,141]. Whether ganglioside-seronegative patients have other antibodies relevant to GBS is currently unclear. Studies have occasionally, though inconsistently, reported antibodies reactive to myelin proteins, such as P0, P2 and PMP22, in GBS[227] but their disease specificity and pathological relevance have not been well established. Unbiased screening of GBS sera against teased nerves or myelinating cultures suggests that other targets may exist[228].

In the past 5 years, antibodies to nodal (neurofascin 186) and paranodal (neurofascin 155, contactin-associated protein 1 or contactin 1) cell-adhesion molecules have been described in patients initially diagnosed with GBS[229,230]. However, detection of these antibodies is now generally accepted to define a separate group of diseases, collectively termed autoimmune nodopathies[8,9]. Autoimmune nodopathies may be clinically and electrophysiologically indistinguishable from GBS in early stages but require distinct therapeutic approaches and have a different prognosis. Thus, testing for these antibodies has practical implications in suspected GBS.

involving swelling of peripheral nerve myelin before infiltration of inflammatory cells. The authors concluded that neurological dysfunction occurred before lymphocytes and macrophages appeared, and speculated that these cells might instead be involved in the reparative process[71]. This view was disputed by other authors who described multifocal lymphocytic infiltration and/or macrophage-mediated segmental demyelination as the primary pathological mechanism in GBS; however, these discrepancies may be due to differentially timed tissue sampling[71]. In direct support of this, subsequent human autopsy studies showed that complement deposition on degenerating myelin in the early stages of the disease was replaced by increased numbers of cytotoxic $CD8^+$ T cells, and upregulation of MHC class I on Schwann cells, at later time points[72].

More recently, it has been argued that the inflammatory demyelination and endoneurial oedema that can be detected histologically in the proximal spinal nerves of patients with GBS may lead, at least in some patients with severe disease, to a constriction of transperineural vessels, endoneurial ischaemia and distal axonal degeneration[93]. How this oedema appears is unknown but the cascade of events suggests that it appears upon antibody-mediated complement activation at the injury site. This also suggests that finding primary demyelination rather than axonal degeneration may not reflect pathologically distinct GBS subtypes but rather be a consequence of other, diverse factors. The electrophysiological heterogeneity of GBS, in which demyelinating, axonal and mixed patterns appear in patients with similar clinical phenotypes and triggers, would also argue in favour of this view.

Cytokines and cellular immunity

Cellular immune effectors and cytokines have also been implicated in the pathology of GBS. In the initial stages of clinical disease, reduced numbers of regulatory T (T_{reg}) cells can be observed in peripheral blood[94]. Furthermore, drugs that supress T cell and T_{reg} function (such as tacrolimus[95] or the more specifically T_{reg}-targeting anti-CD25 antibody–drug conjugate camidanlumab tesirine[96]) or inhibit the activity of T cell immune checkpoints (such as ipilimumab and nivolumab)[97] are associated with GBS-like presentations in some patients. In addition, the peripheral blood neutrophil-to-lymphocyte ratio is elevated early in the disease course of GBS, is higher in patients with poor outcomes, and normalizes after treatment. Although this may simply be due to the preceding infection, it suggests an initial role for innate immunity in GBS[98]. In keeping with this, dendritic cells from patients with GBS and culture-proven *C. jejuni* infection who were in remission show enhanced responsiveness to *C. jejuni* LOS, which suggests a constitutively enhanced response to *C. jejuni* LOS that would be the activation signal promoting a T cell-independent autoantibody response directed towards GM1 (structurally similar to LOS)[78]. Furthermore, peripheral blood mononuclear cells from patients with GBS secrete increased levels of TNF and IL-1β following exposure to lipopolysaccharide[99] and polymorphisms in some innate immunity receptors (such as Toll-like receptor 4 (TLR4)) have been linked to the development of GBS[100]. Experimental studies in rats show that intraneural TNF injection induces macrophage-mediated demyelination[101], mimicking ultrastructural findings in GBS biopsy samples, where macrophages can be observed specifically digesting either axons or their surrounding myelin sheath[102]. In further support of a role for TNF, overall serum levels are higher in patients with GBS than in healthy controls[103]. Furthermore, a specific polymorphism in the *TNF* gene is associated with an increased risk of GBS[104] and TNF blockers have been linked to the development of demyelinating neuropathies, including GBS-like presentations[105].

The role of T cells in the pathogenesis of GBS is less clear. CD4 and CD8 T cell infiltrates are found in peripheral nerves of most patients with GBS (20 of 22 patients in one study)[106–109], and a range of cytokines related to T helper 1 (T_H1), T_H2 and T_H17, such as IL-6, IL-17A and IL-22, are elevated in the CSF and serum of patients with GBS. However, whether T cells and/or their associated cytokines have a direct effect on axonal and myelin integrity is unknown[110]. Autoreactive CD4 T cells with cytotoxic phenotypes and CD8 T cells targeting myelin antigens have been described in patients with GBS classified as having the 'demyelinating' subtype on electrophysiological grounds[72,103,111]. Moreover, GBS-associated ganglioside antibodies are typically IgG and show evidence of somatic hypermutation[112], suggesting an antigen-driven, germinal centre reaction, in which follicular $CD4^+$ T cells are involved. At the same time, GBS is usually a monophasic disorder, driven by a transient humoral attack against glycosylated, non-protein antigens in which TLRs, capable of inducing B cell maturation and antigen-driven reactions, particularly in the ageing population, seem to have a prominent role.

All of these observations argue in favour of a T cell-independent humoral response but the exact mechanisms and their relative contributions to disease pathogenesis may vary across GBS variants[111]. Larger immune-cell profiling studies and a better knowledge of predisposing genetic factors could help clarify the pathophysiological pathways leading to the appearance of anti-neural antibodies in patients with GBS.

Animal models

The use of animal disease models has also advanced the understanding of GBS pathogenesis. Experimental autoimmune neuritis was first described based on active immunization of rabbits with homogenized peripheral nervous tissue[113]. Subsequent modifications used specific peripheral nerve fractions (for example, myelin) or defined antigens (for example, purified P2 myelin protein) as the immunogen. Later, driven by serological findings in patients with GBS, active immunization with gangliosides[114] or passive transfer of ganglioside antibodies[115] was used to induce peripheral nerve dysfunction in experimental animals. GBS itself is almost never induced via such mechanisms[116,117], but these models nevertheless provide useful insights into the neuroimmune interactions and the process of autoimmune demyelination of the peripheral nerves that may be relevant to GBS pathogenesis. Experimental autoimmune neuritis models demonstrated that the response to some immunogens induces targeted demyelination of the peripheral nervous system, whereas other immunogens additionally affect the central nervous system[113]. Furthermore, adoptive T cell transfer[118,119] or passive transfer of serum alone[120,121] from experimental autoimmune neuritis animals into non-immunized animals was shown to recapitulate aspects of the disorder, showing that both humoral and cellular immunity could contribute to neuropathology within the same disease. Ganglioside models revealed that systemic ganglioside expression is crucial in maintaining self-tolerance[122] and that neuronal ganglioside expression has a dominant role in clearing ganglioside antibodies from the circulation via endocytosis[89]. In addition, these models demonstrated the complement-dependent[115] and complement-independent[123] downstream effects of ganglioside antibodies and provided a strong rationale for subsequent clinical trials of complement inhibition in GBS.

Diagnosis, screening and prevention

Clinical patterns

The clinical hallmark of typical motor-sensory and motor forms of GBS is a progressive flaccid limb paresis and areflexia. Sensory loss is also present in the predominant motor-sensory form but absent in the motor form. The most severely affected patients also develop respiratory failure and autonomic dysfunction[124]. By contrast, the clinical hallmark of MFS — the most recognizable variant of GBS — is a triad of ophthalmoplegia, ataxia and areflexia[125]. In other clinical presentations, patterns of weakness affect specific body regions, such as paraparetic forms with only weakness of the legs. Despite their heterogeneous presentations, these 'regional' variants of GBS are typically an incomplete clinical expression of typical GBS and/or MFS[126]. For example, patients with Bickerstaff brainstem encephalitis have central nervous system features that include altered levels of consciousness and extensor plantar responses, overlapping with MFS and sometimes motor-sensory GBS[126] (Fig. 2 and Table 1). The reasons underlying the different clinical manifestations are, to some extent, associated with the type of anti-glycolipid antibodies in a patient as well as their antecedent illness. For instance, patients with *C. jejuni* infection have anti-GM1 and anti-GD1a antibodies and are susceptible to developing motor and axonal forms of GBS, whereas patients who have IgG against GQ1b develop clinical features of MFS[125,126].

Diagnostic criteria

To facilitate an early diagnosis of GBS, several diagnostic criteria have been established. Previously, criteria were developed by the US National Institute of Neurological Disorders and Stroke (1990) and the Brighton Collaboration GBS Working Group (2011) in response to post-vaccination surveillance[127,128]. An international consensus guideline on GBS management that was easy to implement in a global setting was developed in response to the Zika virus outbreak (2015–2016)[129]. Most recently, the EAN and PNS together established guidelines for the diagnosis and treatment of GBS (2023)[8,9] (Box 2).

In all diagnostic criteria, GBS is regarded as a clinical diagnosis supported by additional investigations. Patient characteristics are classified as either required or supportive for the diagnosis or inconsistent with the diagnosis. The presence of symmetrical flaccid limb paresis with hyporeflexia or areflexia is required to establish a diagnosis of GBS with a high degree of certainty. The EAN–PNS guidelines added "progressive worsening of not more than 4 weeks" as a diagnostic requirement[8,9]. This criterion had previously been considered supportive of a diagnosis of GBS. However, robust evidence from IGOS and other cohort studies suggests that the vast majority of patients with GBS reach the plateau phase by 2 weeks (80–97%) and almost all do so by 4 weeks (97–99.8%)[4,130]. It is also becoming increasingly apparent that, when there is progression of weakness beyond 4 weeks, a diagnosis of acute-onset chronic inflammatory demyelinating polyneuropathy (CIDP) or autoimmune nodopathy should be considered[131].

Other clinical features in support of a diagnosis of GBS include relative symmetry in neurological symptoms and signs, sensory symptoms and signs that are mild or absent, involvement of cranial nerves (in particular bilateral facial palsy), autonomic dysfunction, respiratory muscle weakness, and pain[8,129]. The sensitivity of these clinical characteristics varies between patient cohorts and depends on geographical region, timing and disease severity. Overall, in the IGOS-1000 cohort, the frequencies at study entry were 99% for bilateral involvement, 59% for sensory symptoms, 50% for cranial nerve involvement, 25% for autonomic dysfunction and 55% for pain[4].

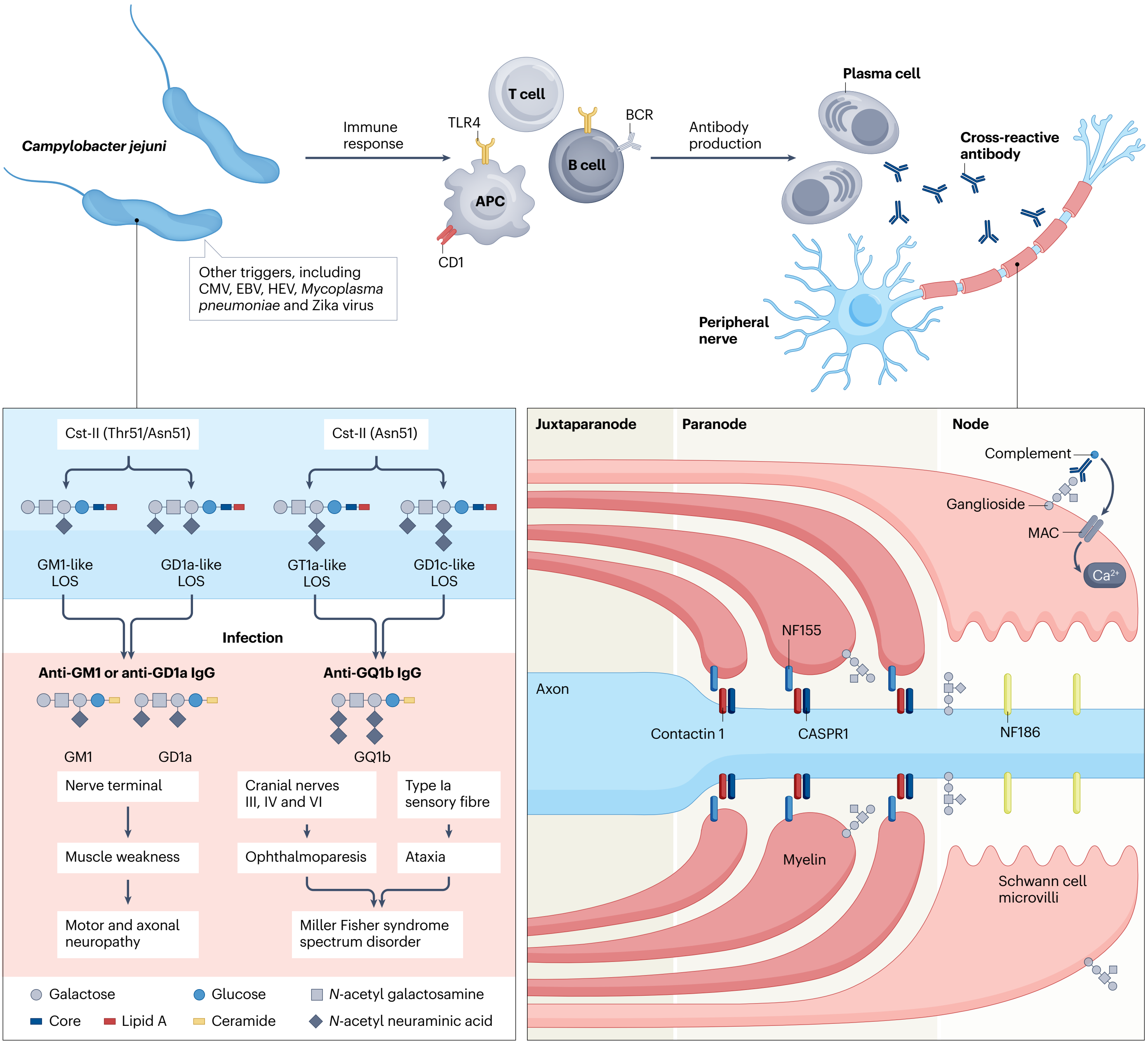
By contrast, the presence of asymmetrical weakness, fever, upper motor neuron signs, abdominal pain or vomiting, and nystagmus (involuntary, rapid eye movement) is unusual in GBS. In addition, patients with relatively mild limb weakness do not typically develop severe respiratory impairment nor do they exhibit significant sensory signs. Disease progression that is short (<24 h) or prolonged (>4 weeks) make GBS unlikely[8,129]. Other differential diagnoses should be considered in these scenarios (Table 2).

Two-thirds of patients with GBS have a history of an antecedent event within 6 weeks before the onset of neurological symptoms. Although absent from previous diagnostic criteria, this is included in the EAN–PNS guidelines as supporting a diagnosis of GBS. These antecedent events include infections (especially diarrhoeal and respiratory illnesses) or, rarely, surgery, malignancy, trauma and certain vaccinations[28,32,33,64]. Early diagnosis is required for a timely start of monitoring and treatment of GBS but may be complicated by the rarity of disease, the diversity in clinical presentation and the extensive differential diagnosis (Table 2). Misdiagnosis of GBS, while the patient is in the progressive phase, may result in respiratory or autonomic failure. One study showed that, in preschool children with GBS, two-thirds were initially diagnosed with infection, synovitis or musculoskeletal disease, which in some resulted in emergency intubation or death[132]. In any clinical setting, the threshold for suspecting GBS should be low in individuals with symptoms and/or signs of peripheral nerve dysfunction within the context of relevant preceding events and triggers[132].

Laboratory findings

In conjunction with clinical features, certain investigations support the diagnosis of GBS or can help to rule out potential mimics. These are typically only accessible in a hospital setting.

CSF analysis. In patients with GBS, the characteristic CSF finding is albumino-cytological dissociation, in which CSF protein levels are raised (>0.45 g/l) with a normal white cell count (usually <5 cells/µl,



rarely 5–50 cells/µl (refs. 4,8,9,130)). However, patients can have normal CSF protein levels in the first week of disease onset, which does not exclude the diagnosis of GBS[130,133]. CSF protein levels also depend on the timing of the lumbar puncture during the disease course. In IGOS and a previous cohort, increased CSF protein levels were found in <50% of patients at 1 day and in >80% at >1 week after onset of weakness[130,133]. Importantly, a finding of CSF pleocytosis (>50 cells/µl) in general is atypical and should alert clinicians to consider other differential diagnoses including infectious polyradiculitis or an infiltrative process such as malignant leptomeningeal metastasis[125].

Neurophysiology studies. NCS are helpful in establishing the presence of a poly(radiculo)neuropathy, supporting a diagnosis of GBS. Electrophysiological findings change over time and the information that these studies provide varies depending on the timing of the assessment. When performed at the initial stages of the disease (within the first 2 weeks of disease onset), standard parameters, such as nerve conduction velocities and amplitudes, may appear normal. However, late responses can be abnormal at the early disease stage, including an absent H reflex and F waves, as well as the presence of abundant A waves[134,135]. In GBS, an absent H reflex has >95% sensitivity, making GBS highly unlikely if the H reflex is present[8,9]. Neurophysiological features that are more specific for GBS tend to appear later (after 4 weeks) and include prolonged distal compound muscle action potential duration, indirect discharges (A waves) and a sural-sparing pattern (specificity 91–98%), which is demonstrated by a normal sural sensory nerve action

Fig. 4 | Pathophysiological mechanism of GBS. The pathophysiology of Guillain–Barré syndrome (GBS) starts with a precedent infection (or other preceding event) that triggers an immune response in susceptible patients and results in damage to peripheral nerve myelin sheaths, axons or both, causing the subsequent progressive neurological deficits. In this immune response, autoantibodies targeting gangliosides and other unknown antigens are the specific effector mechanisms that bridge the immune response against the infectious trigger and the autoimmune attack to the nerve. Anti-ganglioside antibodies, which are transiently present in the blood of patients with GBS, are triggered by an infection with *Campylobacter jejuni*, the predominant type of preceding infection in GBS worldwide. Molecular mimicry between carbohydrates of the lipo-oligosaccharide (LOS) of *Campylobacter* and gangliosides of human nerves results in a cross-reactive antibody response. The factors leading to autoantibody production are unknown, but innate immune cells and receptors (dendritic cells, Toll-like receptors (TLRs)) and molecules involved in presentation of glycolipid antigens (such as CD1) are thought to be involved in disease pathogenesis. The role of T cells in the pathogenesis of GBS is still undefined, but there is evidence for T cell reactivity to peripheral nerve antigens and involvement of T helper 17 cells and associated cytokines. Autoantibodies are likely secreted by short-lived plasma cells. Biosynthesis of the ganglioside mimics in LOS depends on the activity of the sialyltransferase Cst-II, which differs between strains of *C. jejuni*. LOS from strains with monofunctional (a2,3) sialyltransferase Cst-II (Thr51/Asn51) activity induce antibodies to gangliosides such as GM1 and GD1a, whereas those with bifunctional (a2,3 and a2,8) sialyltransferase Cst-II (Asn51) induce antibodies to gangliosides such as GT1a and GQ1b. The resulting antibody specificity influences the clinical phenotype owing to specific ganglioside enrichment at different peripheral nervous system sites. Antibodies binding the nerves cause local complement activation, resulting in the formation of membrane attack complex (MAC) and subsequent neural damage. *Mycoplasma pneumoniae* may also trigger a cross-reactive antibody response to the glycolipid galactocerebroside on peripheral nerves; the pathogenic mechanisms of action in GBS after preceding viral infections are largely unknown. APC, antigen-presenting cell; BCR, B cell receptor; CASPR1, contactin-associated protein 1; CMV, cytomegalovirus; EBV, Epstein–Barr virus; HEV, hepatitis E virus; NF155, neurofascin 155; NF186, neurofascin 186.

potential but abnormal or absent median and/or ulnar sensory nerve action potentials. This feature appears in all subtypes of GBS[8,9,136].

In the past, several criteria were used to dichotomize the electrodiagnostic patterns of GBS into demyelinating (acute inflammatory demyelinating polyneuropathy) and axonal (acute motor axonal neuropathy or acute motor-sensory axonal neuropathy) subtypes, with the aims of better understanding the underlying pathophysiology and to determine disease prognosis[19,69,137]. This approach may have some merit but it also has limitations. As discussed in the preceding sections, the complex immunological interplay that leads to peripheral nerve damage in GBS can result in both segmental demyelination and axonal damage. Thus, trying to determine a GBS subtype based on a single NCS might be overly simplistic.

Antibodies. Several studies have described the association of GBS with specific antibodies but most are of limited value in routine clinical practice. The exception to this is serological testing for serum IgG against GQ1b, which is reported to be 100% specific for MFS or MFS–GBS overlap syndromes. It is recommended that this test is conducted to facilitate early diagnosis if it is available[138,139]. Testing for antibodies to nodal and paranodal antigens (including contactin 1, contactin 1–contactin-associated protein 1 complex, neurofascin 155, neurofascin 140–neurofascin 186, contactin-associated protein 1) could be considered in patients who have a poor response to treatment or continue to progress or relapse[140]. Depending on the extensiveness of the screening and with current methods, no serum antibodies are detected in 30–80% of patients with GBS[19,22,75–77,82,83,138,139,141].

Neuroimaging. An emerging area of research in immune-mediated neuropathies is imaging of the peripheral nerves and nerve roots with ultrasonography or magnetic resonance imaging (MRI)[142–144]. Currently, neither technique has a role in the routine diagnosis of GBS but they can be considered to support the diagnosis in atypical presentations or to exclude other causes. For instance, MRI of the spine can help rule out spinal cord disease in the paraparetic variant of GBS, whereas a finding of widespread nerve enlargement on imaging is supportive of the differential diagnosis of acute-onset CIDP. Nerve root enhancement on MRI is a frequent finding in GBS but not specific for the diagnosis.

Screening and prevention

Knowing the potential triggers for GBS helps with measures to prevent exposure when possible. For instance, in a study from New Zealand, a countrywide campylobacteriosis control programme that focused on reducing contamination in poultry resulted in a fall in campylobacteriosis as well as a considerable decrease in GBS hospital admissions[145]. However, not all individuals with exposure to infectious agents known to be associated with GBS eventually develop the disease. For instance, only 1 in 1,000 patients with a *C. jejuni* infection develop GBS, making surveillance of GBS in specific infections not cost-effective[43,44]. Other features, such as genetic factors, may result in an increased susceptibility to developing GBS[78].

Management

Recommendations for the management of GBS are largely based on the international consensus report and guidelines from the EAN–PNS[8,9,129].

Prediction of clinical course and outcome

GBS usually has a characteristic monophasic disease course, but the rate of progression and recovery varies considerably between patients. Factors that have previously been associated with poor prognosis are high age, severe disability and weakness in the acute stage, cranial nerve involvement, preceding infections with *C. jejuni* or cytomegalovirus, antibodies to gangliosides GM1 and GD1a, high levels of neurofilament light chain (NfL), genetic polymorphisms (*MBL2* gene) and electrophysiological characteristics (decreased compound muscle action potential amplitude and inexcitable motor nerves)[11,124,141,146–153]. In addition, the response to treatment is variable. For IVIg, this is in part related to its pharmacokinetics as a higher rate of consumption of IgG is associated with poorer recovery in patients with GBS[146,147]. In general, outcome is defined by the combination of factors related to the severity of early immune-mediated nerve damage, treatment and capacity of nerves to regenerate.

The guidelines recommend using validated prognostic models for clinical practice, which are currently based largely on clinical predictors but, in the future, will be combined with blood biomarkers such as presence of autoantibodies or levels of NfL[152,153].

The Erasmus GBS Respiratory Insufficiency Score (EGRIS) is an accurate and validated prognostic model that determines the risk

Box 2 | Diagnostic criteria for motor-sensory GBS based on European Association of Neurology–Peripheral Nerve Society guidelines (2023)

Required features
- Progressive weakness of arms and legs
- Absent or decreased tendon reflexes in affected limbs
- Progressive worsening of ≤4 weeks

Supportive features
- Relative symmetry
- Relatively mild or absent sensory symptoms and signs
- Cranial nerve involvement (especially bilateral facial palsy)
- Autonomic dysfunction
- Respiratory insufficiency (from muscle weakness)
- Pain (muscular or radicular in back or limb)
- Recent history of infection (<6 weeks)

Supportive laboratory tests
- **Cerebrospinal fluid**
  - Increased protein (normal does not exclude) and white cell count $<5\times10^6$/l
- **Serology**
  - Serum anti-GQ1b antibodies in Miller Fisher syndrome
- **Electrodiagnostic**
  - Nerve conduction studies consistent with peripheral neuropathy (can be normal in early disease)

Findings inconsistent with GBS
- Asymmetric weakness (marked and persistent)
- Severe respiratory dysfunction at onset with mild limb weakness
- Predominant sensory signs at onset with mild weakness (paraesthesias often occur)
- Fever at onset
- Sensory level or extensor plantar responses
- Hyper-reflexia (initial hyper-reflexia does not exclude Guillain–Barré syndrome (GBS))
- Bladder and/or bowel dysfunction (does not exclude GBS)
- Abdominal pain or vomiting
- Nystagmus
- Alteration of consciousness (except in Bickerstaff brainstem encephalitis)
- Abnormal routine blood tests
- Cerebrospinal fluid $>50\times10^6$/l mononuclear or polymorphonuclear cells

Adapted with permission from refs. 8,9, Wiley.

of respiratory failure in the first week in individual patients with GBS[154,155] (Table 3). The model predictions are based on three clinical factors determined at hospital admission: time from onset of weakness to admission (days), presence of facial and/or bulbar weakness, and severity of limb weakness defined by the Medical Research Council-sumscore[154]. The EGRIS has been validated in independent cohorts from several Western and Eastern countries, including LMICs. The EGRIS can be calculated via various online tools and is used in many countries (for example, via the IGOS prognosis tool or QxMD). The modified EGRIS (mEGRIS) is a new, simplified version of EGRIS, based on time from onset of weakness to admission, presence of bulbar weakness, and weakness of neck flexion and bilateral hip flexion[156]. The mEGRIS is as accurate as the EGRIS but performs better in patients with mild forms of GBS and has the advantage of predicting respiratory failure at variable time points up to 2 months; however, the mEGRIS still requires external validation. The current EAN–PNS guideline advises the use of a prognostic model to calculate the risk of respiratory failure in individual patients. Predicted risks are helpful for the triage of patients to high-care or intensive care units and to prevent emergency intubations. The risk of not being able to walk at 4 weeks, 3 months and 6 months can be calculated at hospital admission using the modified Erasmus GBS Outcome Score (mEGOS), which is based on patient age, presence of preceding diarrhoea and Medical Research Council-sumscore[157–159] (Table 3).

GBS as a medical emergency

In the early progressive stage of disease, GBS should be regarded as a medical emergency. Bulbar weakness is present in ~25% of patients and may cause obstruction of the upper airways and require intubation[4]. Respiratory muscle weakness may result in failure of respiration in ~20% of patients and can be insidious[4]. Autonomic dysfunction, which occurs in ~25% of patients, may cause severe fluctuations in heart rate and blood pressure and result in circulatory failure[4]. All vital functions need to be secured before specific treatment of GBS can be considered. Depending on the presence of risk factors for respiratory failure or autonomic instability, patients may need to be transferred to high-dependency or intensive care units upon arrival to the hospital. In each patient with suspected GBS, it is recommended to monitor the following items during the progressive phase of the disease: forced vital capacity (at least every 4 h), single breath count, maximum inspiratory and expiratory pressure (if possible), heart rate and blood pressure, clinical examination (at least daily), and routine laboratory tests[8,9].

Initial immune treatment

Based on current Cochrane reviews, plasma exchange and IVIg are the only proven effective forms of immune treatment for the acute stage of GBS[68,160–162]. There is no evidence that corticosteroids or other immune treatments are effective in GBS[68,162]. Both IVIg and plasma exchange aim to prevent further damage of the peripheral nerves by the immune system and are considered equally effective in GBS[163–173]. In general, there is no preference for plasma exchange or IVIg and the decision in practice will largely depend on local availability and costs. When indicated, treatment should be started as soon as possible ('time is nerve'). The recommended regimen for plasma exchange is four to five exchanges over 1–2 weeks but two exchanges are also effective in patients with mild forms who are unable to run but can still walk[161,167,174]. IVIg is administered over 5 consecutive days[8,9]. The most common

Table 2 | Differential diagnosis of Guillain–Barré syndrome

Diagnosis	Clinical features that distinguish from Guillain–Barré syndrome	Supportive tests
Brain		
Encephalitis	Drowsiness, seizures	CSF: pleocytosis MRI brain: hyperintense lesions EEG: slowing with or without epileptiform discharges
Brainstem stroke	Hyperacute sudden onset	MRI or MRA: corresponding infarct and vascular occlusion
Spinal cord		
Transverse myelitis	Sensory level, brisk reflexes	MRI spine: hyperintense lesion
Malignant infiltration	Cauda equina syndrome	CSF: malignant cells MRI spine: enhancing lesions
Anterior horn cell		
Polio, enterovirus 71 or enterovirus D58 infection	Fever	CSF: pleocytosis NCS: normal SNAPs Microbiology: presence of virus
Plexus		
Neuralgic amyotrophy	Asymmetry, pain, findings limited to nerves affected	MRI brachial plexus: nerve enhancement NCS: abnormalities restricted to affected nerves
Nerve roots		
CMV and HIV radiculitis	Subacute presentation	Microbiology: HIV and CMV serology CSF: pleocytosis NCS: abnormal H reflex and F waves
Peripheral nerves		
CIDP	Subacute presentation and relapsing–remitting pattern	CSF: albumin-cytological dissociation NCS: demyelinating neuropathy Neuroimaging: widespread enlarged nerve roots
Porphyria	Family history, concomitant psychiatric symptoms and abdominal pain	Biochemistry: raised urinary porphobilinogen NCS: axonal neuropathy
Lyme disease or other tick-borne disease	History of exposure, characteristic rash (erythema migrans in Lyme)	Microbiology: Lyme disease and tick serology NCS: axonal neuropathy
Thiamine deficiency	Precipitating factors (for example, hyperemesis gravidarum, alcoholism, nutritional deficiency), other neurological features (for example, Wernicke encephalopathy)	Biochemistry: abnormal thiamine levels and erythrocyte transketolase activity NCS: axonal neuropathy
Diphtheria	Laryngeal infection	Microbiology: isolation of *Corynebacterium diphtheriae* on cultures CSF: albumin-cytological dissociation NCS: demyelinating neuropathy
Critical illness polyneuropathy	Prolonged illness or ventilation	NCS: axonal neuropathy with or without myopathic features on EMG
Metabolic or electrolyte imbalance	Precipitating factors	Biochemistry: low serum levels of specific electrolyte
Neuromuscular junction		
Myasthenia gravis	Fatigable weakness, relapsing–remitting	Immunology: acetylcholine receptor antibodies NCS: decremental response on repetitive nerve stimulation
Botulism	Rapid progression, pupillary abnormalities, dysautonomia, descending paralysis	Microbiology: presence of botulism toxin NCS: incremental rise on rapid repetitive nerve stimulation
Lambert–Eaton syndrome	Proximal weakness, depressed tendon reflexes, autonomic changes	Immunology: antibodies against voltage-gated calcium channels NCS: post-tetanic facilitation on repetitive nerve stimulation
Muscle		
Inflammatory myositis	Proximal weakness, normal reflexes and sensation	Biochemistry: raised creatine kinase NCS: normal sensory studies and features on EMG
Critical illness myopathy	Prolonged illness or ventilation	NCS and EMG: myopathic features and normal sensory studies unless overlap with neuropathy

Table 2 (continued) | Differential diagnosis of Guillain–Barré syndrome

Diagnosis	Clinical features that distinguish from Guillain–Barré syndrome	Supportive tests
Muscle (continued)		
Hypokalaemic periodic paralysis	Transient weakness, family history, precipitating factors (for example, fasting, exercise, carbohydrate meals)	Biochemistry: low potassium level Genetic studies: mutation in known genes NCS: abnormal exercise test
Colchicine-induced (neuro)myopathy	Proximal weakness, normal reflexes and sensation	Biochemistry: raised creatinine kinase At risk: renal insufficiency, statins, tacrolimus NCS: myopathic features on EMG
Functional disorder	Inconsistent, variable presentation	Psychological evaluation

CIDP, chronic inflammatory demyelinating polyneuropathy; CMV, cytomegalovirus; CSF, cerebrospinal fluid; EEG, electroencephalogram; EMG, electromyography; MRA, magnetic resonance angiography; NCS, nerve conduction studies; SNAPs, sensory nerve action potentials. Includes data from ref. 124.

adverse effects of plasma exchange are catheter-related events, paraesthesia and blood pressure changes, and of IVIg are flu-like symptoms, urticaria and other skin manifestations[169,173–175].

The decision to start immune treatment in GBS depends on the clinical severity, rate of progression and time from onset of weakness (Table 4). The GBS disability score (GBS-DS) is frequently used to classify clinical severity, ranging from 0 (a healthy state) to 6 (death), and is largely based on the ability of a patient to walk and breathe. Patients who are unable to walk independently (GBS-DS ≥3) and are within 4 weeks of onset of weakness require treatment with either plasma exchange or IVIg[176]. In addition, patients who are still able to walk independently (GBS-DS ≤2) but their disease is rapidly progressing, develop swallowing problems or have a high risk of requiring ventilator support, also have an indication to be treated. Additionally, in patients with a milder form of GBS who are still able to walk independently but unable to run (GBS-DS ≤2), treatment could be considered within 2 weeks of onset of weakness[8,9]. A previous observational comparative study in these patients found no better outcome after one course of IVIg compared with no treatment[177]. No treatment is required in patients with even milder forms or who are stable or recovering and within 2–4 weeks after onset[8,9].

There is no evidence that patients with GBS should be treated differently depending on their particular clinical variant or electrophysiological subtype. Randomized trials supporting the use of plasma exchange or IVIg are not available for MFS but it is generally accepted that treatment strategies should follow GBS recommendations, particularly for those patients unable to walk independently. Children with GBS are treated in the same way as adult patients, with either IVIg (in the same regimen recommended for adults) or plasma exchange[8,9,175]. Potential contraindications to plasma exchange are haemodynamic instability and contraindications to IVIg are IgA deficiency and/or previous severe allergic responses.

After initial immune treatment, most patients stabilize or start to improve. Patients unable to walk independently will regain the capacity to walk after IVIg treatment (45%, 70% and 81% of patients at 4 weeks, 3 months and 6 months after IVIg, respectively)[157]. Despite this early improvement, a considerable proportion of patients will have persistent deficits and limitations that affect their quality of life (QoL).

Table 3 | Clinical prognostic models for GBS

	Clinical predictors	Scores	Predicted outcome	Risk range[a]	Refs.
Respiratory failure					
EGRIS	Rate of progression[b] Facial and/or bulbar weakness MRC-sumscore[c]	0–7	Respiratory failure in first week	1–91%	154,155
mEGRIS	Rate of progression[b] Bulbar weakness Weakness neck flexion MRC hip flexion	0–32	Respiratory failure in first 2 months	0–100%	156
EGRIS-Kids	Age Cranial nerve involvement GBS disability score[d]	1–9	Respiratory failure in first week	4–50%	203
Inability to walk independently					
EGOS	Age Preceding diarrhoea GBS disability score[d]	1–7	Inability to walk at 6 months	1–83%	204
mEGOS (day 0)	Age Preceding diarrhoea MRC-sumscore[c]	0–9	Inability to walk at 4 weeks, 3 months and 6 months	3–89%	157–159
mEGOS (day 7)	Age Preceding diarrhoea MRC-sumscore[c]	0–12	Inability to walk at 4 weeks, 3 months and 6 months	2–92%	157–159

EGOS, Erasmus GBS Outcome Score; EGRIS, Erasmus GBS Respiratory Insufficiency Score; EGRIS-Kids, EGRIS in children (<16 years); GBS, Guillain–Barré syndrome; mEGOS, modified EGOS (at admission or at day 7 of admission); mEGRIS, modified EGRIS; MRC, Medical Research Council. [a]Risk range describes the range of changes of outcome event predicted by the prognostic model. [b]Rate of progression describes the days between onset of weakness and hospital admission. [c]The MRC-sumscore is the sum of MRC scores for weakness of 6 bilateral limb muscle groups (ranging from 0 (full tetraparalysis) to 60 (normal strength)). [d]The GBS disability score ranges from 0 (healthy) to 6 (death).

Insufficient response to initial treatment

Patients may progress during or shortly after initial treatment with plasma exchange or IVIg or show no signs of clinical response. IGOS showed that, in current practice, these patients are sometimes given a second course of treatment[178,179]. Currently, there is no evidence that additional treatment is effective in this situation. A second course of IVIg in patients with poor prognosis has no beneficial effect on recovery but increases the risk of severe adverse effects, in particular thromboembolic events[8,9,178,180]. It is also not recommended to start

Table 4 | Recommendations for immune treatment of GBS

Disease stage and severity	GBS-DS	Timing[a]	Plasma exchange	IVIg
Acute				
Unable to walk	GBS-DS ≥3	<4 weeks	4–5 exchanges in 1–2 weeks	Over 5 days
Progression or poor prognosis[b]	GBS-DS <3	<4 weeks	4–5 exchanges in 1–2 weeks	Over 5 days
Unable to run (consider)[c]	GBS-DS 2	<2 weeks	2 exchanges in 1 week	Over 5 days
Unable to run but stable	GBS-DS 2	2–4 weeks	None	None
Very mild	GBS-DS 1	<2 weeks	None	None
Follow-up				
Treatment-related fluctuation[d]	NA	NA	4–5 exchanges in 1–2 weeks	Over 5 days
No response	NA	NA	No second course	No second course

Recommendation largely based on good practice points in 2023 European Association of Neurology and Peripheral Nerve Society Guillain–Barré syndrome (GBS) guidelines[8,9]. Either plasma exchange or intravenous immunoglobulins (IVIg) may be provided as treatment for patients with GBS. GBS-DS, GBS disability score; NA, not applicable. [a]Timing describes time from onset of weakness. [b]Patients who are able to walk but show rapid progression have a high risk of respiratory failure, difficulties swallowing or other poor prognostic factors. [c]Treatment could be considered for patients who are able to walk but unable to run and are within 2 weeks of onset of weakness. [d]Treatment-related fluctuation is defined as an initial clinical recovery or stabilization after treatment, followed by a secondary deterioration.

plasma exchange in patients first treated with IVIg, or vice versa[8,9,169]. The absence of an early clinical response in the first days or weeks also does not exclude the possibility that there is an ongoing therapeutic effect. In patients who continue to progress for >4 weeks after onset of weakness, an additional diagnostic work-up needs to be considered to exclude other causes, in particular, testing for serum antibodies to paranodal and nodal proteins to exclude an autoimmune nodopathy that requires alternative treatments[8,9].

A lack of response needs to be distinguished from a setting in which a patient shows initial clinical stabilization or recovery after treatment, but this is then followed by a secondary deterioration. Such a treatment-related fluctuation (TRF) may occur when there is prolonged disease activity that outlasts the duration of effect of the initial treatment. Therefore, the recommendation is to treat a TRF with a second course of the same treatment (either IVIg or plasma exchange)[8,9]. Some patients with GBS may have up to two TRFs, which usually occur within 8 weeks of onset of weakness. In patients with ≥3 TRFs or a deterioration after 8 weeks, the diagnosis of CIDP should be considered[8,9,131,181]. Relapses of GBS may occur years after the first episode and are treated similarly to an initial diagnosis.

Prevention of complications

Patients with GBS are prone to developing complications. Monitoring and preventive measures are important as most complications can be either prevented or treated[129]. Regular communication with patients is essential, especially when admitted to the intensive care unit, to be able to monitor the development of pain, hallucinations, anxiety, and depression and to provide moral support. Daily clinical examinations are required to prevent choking in patients with bulbar palsy, corneal ulceration in patients with facial palsy, cardiac arrhythmias and urine retention in patients with autonomic dysfunction, thromboembolic complications (including deep venous thrombosis and pulmonary embolism), constipation, contractures and pressure ulcerations in immobile patients, and hospital-acquired infections and hyponatraemia in all patients[8,9,129]. Pain is a frequent complication of GBS in all stages of the disease and may even be the presenting symptom before weakness. Various forms of pain can occur, including pain of radicular, neuropathic, myogenic or tendomyogenic origin. Pain treatment depends on its cause, type and intensity. First-choice treatments for neuropathic pain in GBS are gabapentinoids (gabapentin or pregabalin) or tricyclic antidepressants (after exclusion of arrhythmia); second-choice treatments include carbamazepine and serotonin–noradrenaline reuptake inhibitors (duloxetine and venlafaxine)[8,9]. It is recommended to start physiotherapy, occupational therapy, speech therapy, dietitian consultations and rehabilitation treatment as early as possible during the hospital stay[8,9]. Complications are best managed by a multidisciplinary team. It is also recommended to inform all patients and their relatives of the support they can acquire via the GBS-CIDP Foundation International and local patient representative organizations.

Quality of life

Patients with GBS experience various physical, mental and psychosocial complications in the acute and later stages of disease that are often overlooked and may have a negative impact on their QoL[182]. Fatigue is an important complication of GBS, especially during the recovery phase, and may occur in up to 60% of patients[183–186]. Although studies evaluating psychiatric symptoms of GBS are limited, stress, anxiety, depression, psychosis and sleep disturbance have been frequently reported with variable proportions[187] (Table 5). These symptoms usually start during the acute phase of GBS and can occur irrespective of the severity of motor weakness and physical recovery[188]. If these symptoms are left unrecognized or untreated in the acute stage, they can lead to long-term psychiatric problems[187].

Residual symptoms have a widespread impact on daily living, work and social activities, which can last for several years after disease onset[188–190]. The main reasons reported by patients for associated changes are lack of strength, sensory disturbances and psychological consequences such as depression, apathy or impaired concentration[191]. Several instruments are used to measure different dimensions of physical and psychosocial functioning and QoL in patients with GBS. The Inflammatory Rasch-built Overall Disability Scale and Fatigue Severity Scale or Rasch-built modified Fatigue Severity Scale are widely used to capture activity and social participation limitations and fatigue[192]. To measure general health status and QoL, the 36-item Short Form[193,194], Sickness Impact Profile and Nottingham Health Profile are mostly used[193].

The average time required for any improvement of health-related QoL ranges from 6 months to 6 years and the most prominent improvement has been reported within the first year of onset of GBS[188,191,193]. Female sex, age >50 years and treatment in intensive care are associated with poor long-term QoL[188]. The correlation between disease severity and muscle weakness in the acute phase of GBS and long-term QoL are debatable and yet to be confirmed[195]. Wide variations in study designs, follow-up time points and outcome measurement instruments cause difficulties in comparing or combining the findings of different studies. The effect of sociodemographic and clinical determinants on changes in QoL scores over time with the progression and recovery of GBS has rarely been explored. In addition, most of these studies were conducted in the Northern hemisphere, mostly in high-income countries. Considering the differences in socioeconomic conditions or geographical variations in disease presentation, subtype and severity, these findings might not be representative of patients from LMICs or patients from the Southern hemisphere, which merits further evaluation.

Outlook

Despite the progress made, several major outstanding research questions and challenges in the understanding and management of GBS remain (Box 3).

Epidemiology and pathogenesis

In ~50% of patients with GBS, the antecedent event that triggers the onset of disease is still unidentified[28]. Detecting a preceding infection can be challenging because of the delay in onset of neurological deficits and time of testing, especially for studies based on culturing or PCR.

Table 5 | Psychosocial complications and impact on quality of life of patients with Guillain–Barré syndrome

Variables	Percentage (follow-up time points)	Refs.
Depression	18% (1–14 years)	188
Anxiety	82% (during patient stay in ICU)	205
	21–26% (6 months to 14 years)	188,206
Stress	17% (1–14 years)	188
Reactive psychosis	25% (during patient stay in ICU)	205
Sleep disturbances	50% (10 days after admission to 2 weeks)	207,208
Fatigue (FSS)	12–39% (admission to discharge)[a]	209
	42–60% (2–20 years)[b]	185,209,210
	30–37% (3–6 years)	185,190,191
Changes in daily activities	43% (6 months after rehabilitation)	189,211
	24% (2 years)	211
	13–44% (3–6 years)	190,191
Changes in work	20–38% (2–6 years)	190,191,211
	13–44% (3–6 years)	190,191
	19% (3–6 years)	191
Sexual problems	19% (3–6 years)	190,191

FSS, Fatigue Severity Scale; ICU, intensive care unit. [a]FSS ≥ 4. [b]FSS ≥ 5.

Serology and high-throughput screening arrays are likely to be more sensitive at identifying new infections triggering GBS. Surveillance studies of GBS following vaccinations are important for vaccine safety but also because incident cases may undermine vaccination campaigns, even in the absence of an association.

Autoantibodies to many peripheral nerve targets have been reported in patients with GBS, but the role of most antibodies in pathogenesis is still unknown. In 30–60% of patients, antibodies to specific targets are not found, especially in patients with preceding viral infections and demyelinating subtypes of GBS[11,76]. Possible targets are cryptic or complex antigens that may be only present in (live) neurons and, alternatively or in addition, antibodies may be removed from the serum by antigen-mediated clearance[87,88]. Additionally, the role of autoreactive T cells in GBS pathogenesis needs to be established.

Another important outstanding question is whether inherent risk factors for developing GBS exist, such as genetic susceptibility, and if so, how this relates to preceding infections. The general lifetime risk of developing GBS is ~0.1%, but the risk of relapse in individuals with a previous GBS episode is 2–5%, which may indicate an increased susceptibility; however, risk factors for relapse have not been identified[11]. There is no evidence of an increased risk of GBS in families or in association with other autoimmune diseases but studies on genetic polymorphisms with sufficient power are needed.

Diagnosis

The recent EAN–PNS diagnostic criteria require validation in extensive cohorts of patients with GBS and its mimics[8,9]. GBS-like syndromes can occur in relation to treatment with checkpoint inhibitors, malignancies or para-infectious conditions that have a distinct pathogenesis and require a different treatment compared with GBS[97]. Furthermore, not all patients with GBS will be identified by the clinical criteria owing to its d iverse presentation. For example, some patients with GBS may have persistent reflexes, especially those with motor forms. Specific biomarkers to support the diagnosis would be helpful, even if they are only associated with disease in a subgroup of patients. Antibodies to GQ1b are highly specific for MFS or MFS–GBS overlap forms but are only found in 5% of patients within the GBS spectrum. Other antibodies or patterns of antibody activity may identify other small subgroups of GBS. For the diagnostic work-up, early and distinctive features in the electrophysiology of GBS are more relevant than criteria for demyelinating and axonal subtypes. Previous studies in IGOS showed the extensive variation in the use of electrophysiology protocols, reference values and criteria in current practice, which influences the diagnosis and subtyping, highlighting the need for standardization[196–198]. Machine learning will likely support future diagnostic pattern recognition and clinical decision-making.

Treatment and prediction of outcome

Treatment of GBS has remained unchanged since the initial use of plasma exchange and IVIg in the 1980s and early 1990s. This stagnation is remarkable, considering the unmet need for better treatments in GBS and the major progress made in the treatment of other autoimmune or inflammatory diseases in the past decades. The efficacy of new treatments based on enzymatic cleavage of IgG or inhibition of complement is currently being evaluated in GBS[199,200]. Cost-effectiveness studies are important considering the high costs of new treatments that may be used only in a subgroup of patients. Studies in IGOS showed that, in the absence of implemented evidence-based protocols, the current

Box 3 | Key topics and objectives for future research in GBS

Epidemiology and pathogenesis

Preceding events in case-controlled studies

- Infections other than established types, especially respiratory viruses and multiple infections
- Surveillance and registry studies for vaccine safety, especially new vaccines
- Potential other triggers, especially surgery, trauma, malignancy and treatments

Pathogenesis

- Antibodies to known targets (glycolipid complexes), new targets (including proteins and cryptic antigens), their effects in relation to preceding events, clinical course, treatment response and outcome
- Role of genetic polymorphisms, T cells and complement in disease susceptibility and severity
- Pathophysiology of axonal versus myelin injury, nerve recovery, residual pain and fatigue

Diagnosis

- Validation of current diagnostic criteria in real-world patient cohorts including Guillain–Barré syndrome (GBS) and its mimics
- Biomarkers for early diagnosis and monitoring of disease activity and treatment efficacy
- Standardization of electrophysiological studies and development and implementation of protocols for diagnostic work-up in clinical practice
- Development of strict case definitions for research (population-based, surveillance, trials)

Treatment and prediction of outcome

- Understanding the role of disease mechanisms and pharmacokinetics in the (variation of) treatment response and clinical outcome
- Improve clinical and patient-reported outcome measures for treatment evaluation and follow-up
- Improve current trial designs and use of real-world databases for treatment evaluation
- Conduct more trials with potential treatments, including inhibitors of complement and IgG
- Develop affordable and effective treatment for patients from underserved regions
- Improve current prognostic models with biomarkers, develop new models for predicting longer-term outcomes, use prognostic models to personalize treatment

Global health and international collaborations

- Preparation for future epidemics and outbreaks that may trigger GBS
- Trustable knowledge hubs for clinicians, scientists and patients, especially from low-income countries
- Sustainable and accessible international databanks and biobanks for surveillance, cohort and treatment studies

treatment of patients is highly variable between hospitals[179]. There is an urgent need to find affordable treatments for patients in LMICs who frequently receive supportive care only. Small-volume plasma exchange is feasible and safe for patients in these countries, but its efficacy needs to be demonstrated[201].

Treatment development will also benefit from improvement in trial design and more sustainable use of data. Primary end points of previous trials were based on changes in the GBS-DS, which does not include arm and cranial functions, or the perspective of patients. Co-variate adjustments and proportional odds analysis will increase the power of the trials[202]. In addition, observational comparative studies based on existing data on the clinical course of GBS will help to optimize the design of trials and provide initial evidence for treatment efficacy. Furthermore, prognostic models are required to develop personalized treatment for this heterogeneous disease. At present, prognostic models, such as EGRIS and modified Erasmus GBS Outcome Score, are based on clinical features but further improvement will come from biomarkers, such as serum albumin, IgG, NfL and antibodies, that were previously shown to have an additional prognostic value.

Global health and international collaborations

Previous infection outbreaks and epidemics, such as with *C. jejuni* and Zika virus, have resulted in an increased incidence of GBS, considerably affecting patients and societies. To be prepared for future infections related to GBS, international collaboration is required to align patient biorepositories and study protocols. To this end, the IGOS was initiated to determine predictors of outcome and to facilitate therapeutic studies and the IGOS databank and biobank are available for future studies on GBS.

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